

1. Problem: drought decides the harvest

- Akmola grows up to 1/4 of Kazakhstan grain; droughts 2010/2012/2019/2021 halved yields.
- Farmer learns post-factum: too late to shift sowing or insure.
- Missing: local forecast, early drought signal, honest insurance.
- Cost of error: -30% to -50% of season revenue.

2. Solution: Qagro decision support

- District-level 2026 yield forecast + decade drought traffic light + index insurance + sowing advice.
- 3 languages RU/KZ/EN. 3 entries: Telegram bot, FastAPI, Streamlit dashboard with PDF/CSV/GeoJSON.
- No black box: 80% interval + SHAP top-3 factors + open risk formula.
- Insurance is decision support, not a tariff.

3. Demo: 30 seconds in the bot

- /start -> language -> Esil district -> spring wheat -> forecast + risk + insurance + PDF.
- Esil/wheat: yellow risk 35.0, payout 1558 KZT/ha. Zerenda/wheat: green 28.5, payout 150 KZT/ha.
- Bot commands: /fields (78 fields), /elevators (12), Share Location -> nearest district (haversine).
- Web: Plotly actual-vs-forecast, Folium risk map, CSV/GeoJSON/PDF export.

4. Data: 1260 rows, honest sources

- Panel akmola_panel.csv: 1260 rows = 10 districts x 21 years (2005-2025) x 6 crops, 0 NaN in yield.
- Yield anchors: BNS stat.gov.kz; 2024-2025 cross-check 6.6 Mt / 7.6 Mt 16.3 c/ha; FAOSTAT QCL + USDA GAIN validation.
- Climate: NASA POWER MERRA-2 + Open-Meteo ERA5 (MJJA tmean/precip/GDD5/heat30/dry_max/ET0/p30_anom + lat/lon/yield_lag1).
- Fields: 78 polygons (60 real OSM via Overpass, ODbL + 18 honest demo 1x2 km); fallback dump Geofabrik Kazakhstan ODbL; official pilot source iaqmola Smart GeoHub.
- Elevators: 12 (Qoldau granaries-map list, coords estimated from OSM, to be verified).
- Sentinel-2: 62 scenes listed via Planetary Computer STAC (no key) + Copernicus Browser / Sentinel Hub for manual NDVI; ndvi_mean=MISSING honest, model works without NDVI.

Sources: Geofabrik ODbL, iaqmola Smart GeoHub, Qoldau granaries-map, Copernicus Browser, Sentinel Hub, Planetary Computer STAC. District = downscaled oblast anchor x agro-zone coef (centroids, not fields).

5. Model: baseline + LightGBM + honest fallback

- Baseline = mean of previous 5 years, same district + same crop (no params, models/baseline.json).
- LightGBM per crop, TimeSeriesSplit(5) + expanding leave-one-year-out 2020-2025, no leakage.
- Forecast 2026: $y_{\text{pred}} + 80\% \text{ interval (OOF residual_std)} + \text{top-3 SHAP values}$.
- 3 LGBM strong: wheat / barley / oats. 3 EXPERIMENTAL baseline: sunflower / rapeseed / flax (LGBM below baseline on hold-out, structural break 2024-2025) -> $y_{\text{pred}} = \text{baseline mean5}$, interval $\times 1.5$, flag experimental:true.

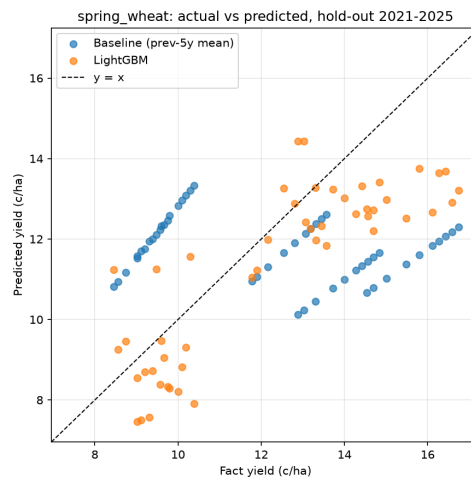


Chart: wheat actual vs predicted (metrics/plots/scatter_spring_wheat.png).

6. Metrics: hold-out 2021-2025, n=50 per crop

- Wheat: baseline MAE 2.68 / R2 -0.26 -> LGBM MAE 1.46 / R2 0.55.
- Barley: 2.78 / -0.25 -> 1.52 / 0.54. Oats: LGBM MAE 1.65 / R2 0.69 (strong).
- Sunflower/rapeseed/flax: LGBM worse than baseline -> honest EXPERIMENTAL baseline fallback (no tuning, metrics fixed in metrics/metrics.json).

Crop	Baseline MAE / R2	LGBM MAE / R2	Status
Spring wheat	2.68 / -0.26	1.46 / 0.55	LGBM strong
Barley	2.78 / -0.25	1.52 / 0.54	LGBM strong
Oats	3.59 / -0.33	1.65 / 0.69	LGBM strong
Sunflower	2.52 / -0.35	2.80 / -0.47	EXPERIMENTAL baseline
Rapeseed	3.70 / -0.85	5.25 / -2.58	EXPERIMENTAL baseline
Flax	1.22 / -0.05	1.65 / -1.07	EXPERIMENTAL baseline

7. Effect: KZT per hectare

- Esil/wheat: mean5 11.46, strike 9.17, P_loss 0.23, expected payout 1558 KZT/ha.
- Zerenda/wheat: P_loss 0.03, payout 150 KZT/ha.
- Early yellow/red July decades -> sowing shift May 12-18 + moisture saving.
- Formula $\max(0, 0.8 \cdot \text{mean5} - y_{\text{pred}}) / 10 \cdot \text{price} \cdot \text{subsidy}$; price 95 000 KZT/t wheat/barley, indicative 70/180/200/220k for oats/sunflower/rapeseed/flax; subsidy 0.8.

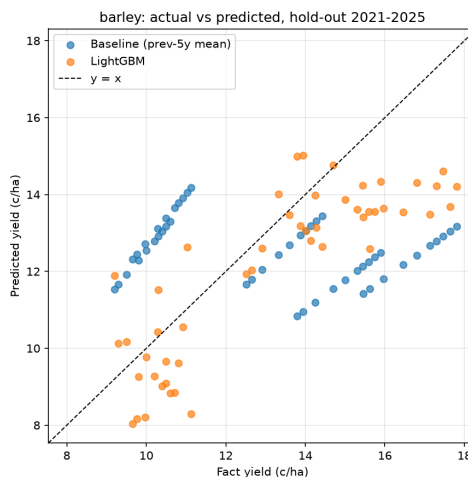


Chart: barley actual vs predicted scatter (metrics/plots/scatter_barley.png).

8. Scale: today and tomorrow

- Today: 10 Akmola districts, 6 crops, 3 languages, Docker (api + bot + web), SQLite 24h cache.
- Geo: 78 fields on Folium map, 12 elevators with haversine nearest search.
- Tomorrow: same pipeline src/fetch_all.py for all Kazakhstan regions; fields instead of centroids.
- Integrations: ERA5-Land soil moisture, Sentinel-2 NDVI per decade as feature + payout trigger, e-APK / Qoldau / insurers.

9. Team

- Team Qagro: Kairbek Ansar - data pipeline / ML / API; Samat Ablayhan (captain) - bot / web / integration / delivery.
- Hackathon 18-21.09.2026: panel -> models -> risks -> insurance -> 3 interfaces -> docs.
- OSS used as pattern only (own code): UniCrop MIT, gsanaev crop-yield MIT, Open-Meteo docs pattern, CropBot MIT - all attributed.

10. Plans: roadmap Q4 2026 -> 2027

- ERA5-Land: soil moisture reanalysis instead of null-soil (hourly soil_moisture_3_9cm already probed).
- Sentinel-2 NDVI: decade vegetation index as model feature and payout trigger (Copernicus Browser / Sentinel Hub pilot).
- e-APK integration: fields / sowings / subsidies + pilots with insurers (index insurance).
- Goal: Akmola -> all Kazakhstan; plot -> field-level decisions.